Resit Exam Calculus A

202001171 (5 EC) / 202001172 (4 EC)

February 16, 2026

18:15–21:15

- This exam consists of Exercises **1–9 for 202001172** and **1–11 for 202001171**.
- You must **motivate** and **justify your answers clearly**. Answers without explanation will not receive credit.
- The use of calculators or any other electronic devices is not permitted.
- A formula sheet is attached to this exam. No other course materials are allowed.

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1. Evaluate the limit:
$$\lim_{x\to\infty}\left(\sqrt{x^2+4x}-x\right).$$

2. The functions f and g, together with their derivatives, take the following values:

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	1	1	2	-1

Find the derivatives with respect to x of the following expressions, evaluated at $x=1$:

(a) $\dfrac{\sqrt{f(x)}}{g(x)}$; (b) $\ln(g(f(x)))$.

3. Evaluate the following integrals:

(a) $\displaystyle\int_e^{\infty}\frac{1}{x\ln x}\,dx$; (b) $\displaystyle\int_0^{\pi^2}\sin(\sqrt{x})\,dx$.

4. Let the function f be given by
$$f(x,y)=\sqrt{x-e^{4y}}.$$
Find an equation of the tangent plane to the graph of f at the point $(5,0,2)$.

5. Let C be the circle with center $(1,-5)$ that is described by the polar equation
$$r^2-2r\cos\theta+ar\sin\theta=-20,$$
where $a\in\mathbb{R}$ is a constant.
   (a) Find a.
   (b) Find the radius of circle C.

6. Show that the limit
$$\lim_{(x,y)\to(0,0)}\frac{2x(x+y)}{x^2+y^2}$$
does **not** exist.

7. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a differentiable function with

$$\frac{\partial f}{\partial x}(x, y) = 2x \quad \text{and} \quad \frac{\partial f}{\partial y}(x, y) = 3y.$$

If $x(t) = t^2$ and $y(t) = \ln t$, compute $\dfrac{d}{dt} f(x(t), y(t))$ for $t > 0$.

8. Consider the function

$$f(x, y) = \frac{1}{3}x^3 + y^2 + 2xy - 6x - 3y + 4.$$

(a) Find all critical points of f.

(b) Determine the absolute minimum of f on the line segment defined by $x = 0$ and $-1 \leq y \leq 1$.

9. Find the solution $y = y(x)$ to the following initial value problem:

$$\begin{cases} x^2 y' + 2xy = \cos x, \quad x > 0, \\ y(\pi) = 1. \end{cases}$$

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10. Find the six complex solutions z (in polar form) of the following equation:

$$z^6 - 2z^3 - 3 = 0.$$

11. Find the solution $y = y(x)$ to the following initial value problem:

$$y'' + 2y' - 3y = 0, \quad y(0) = -1, \quad y'(0) = 0$$

Point distribution:

Exercise	1	2		3		4	5		6	7	8		9	10	11	Total
		a	b	a	b		a	b			a	b				
202001171	4	2	2	2	2	4	2	2	4	4	2	2	4	5	4	45
202001172	4	2	2	2	2	4	2	2	4	4	2	2	4	–	–	36

Exam grade (202001171): $1 + (\text{obtained points}) \times \frac{1}{5}$

Exam grade (202001172): $1 + (\text{obtained points}) \times \frac{1}{4}$

Solutions to Resit Exam Calculus A

202001171 (5 EC) / 202001172 (4 EC)

February 16, 2026

1. **(4 points)** Evaluate the limit:

$$\lim_{x\to\infty}\left(\sqrt{x^2+4x}-x\right).$$

Solution:

This is an indeterminate form of type $\infty-\infty$. We multiply by the conjugate:

$$\begin{aligned}\lim_{x\to\infty}\left(\sqrt{x^2+4x}-x\right) &= \lim_{x\to\infty}\left(\sqrt{x^2+4x}-x\right)\cdot\frac{\sqrt{x^2+4x}+x}{\sqrt{x^2+4x}+x}\\ &= \lim_{x\to\infty}\frac{(x^2+4x)-x^2}{\sqrt{x^2+4x}+x}\\ &= \lim_{x\to\infty}\frac{4x}{\sqrt{x^2+4x}+x}\end{aligned}$$

For $x>0$, divide numerator and denominator by x:

$$=\lim_{x\to\infty}\frac{4}{\sqrt{1+4/x}+1}=\frac{4}{\sqrt{1+0}+1}=\frac{4}{2}=2$$

Answer: $\boxed{2}$

2. **(2+2 = 4 points)** The functions f and g, together with their derivatives, take the following values:

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	1	1	2	-1

Find the derivatives with respect to x of the following expressions, evaluated at $x=1$:

(a) $\dfrac{\sqrt{f(x)}}{g(x)}$; (b) $\ln(g(f(x)))$.

Solution:

(a) Using the quotient rule and chain rule:

$$\begin{aligned}\frac{d}{dx}\left[\frac{\sqrt{f(x)}}{g(x)}\right] &= \frac{g(x)\cdot\frac{d}{dx}[\sqrt{f(x)}]-\sqrt{f(x)}\cdot g'(x)}{[g(x)]^2}\\ &= \frac{g(x)\cdot\frac{f'(x)}{2\sqrt{f(x)}}-\sqrt{f(x)}\cdot g'(x)}{[g(x)]^2}\end{aligned}$$

At $x=1$:

$$=\frac{1\cdot\frac{2}{2\sqrt{1}}-\sqrt{1}\cdot(-1)}{1^2}=\frac{1+1}{1}=2$$

Answer: $\boxed{2}$

(b) Using the chain rule twice:

$$\frac{d}{dx}[\ln(g(f(x)))] = \frac{1}{g(f(x))} \cdot g'(f(x)) \cdot f'(x)$$

At $x = 1$, we have $f(1) = 1$, so:

$$= \frac{1}{g(f(1))} \cdot g'(f(1)) \cdot f'(1) = \frac{1}{g(1)} \cdot g'(1) \cdot f'(1)$$
$$= \frac{1}{1} \cdot (-1) \cdot 2 = -2$$

Answer: $\boxed{-2}$

3. **(2+2 = 4 points)** Evaluate the following integrals:

(a) $\displaystyle\int_e^\infty \frac{1}{x \ln x}\, dx$; (b) $\displaystyle\int_0^{\pi^2} \sin(\sqrt{x})\, dx$.

Solution:

(a) Use substitution: Let $u = \ln x$, then $du = \frac{1}{x}dx$
When $x = e$: $u = 1$; when $x \to \infty$: $u \to \infty$

$$\lim_{a\to\infty} \int_e^a \frac{1}{x \ln x}\, dx = \lim_{b\to\infty} \int_1^b \frac{1}{u}\, du = \lim_{b\to\infty} \ln b - \ln 1 = \infty$$

Answer: $\boxed{\text{The integral diverges}}$

(b) Use substitution: Let $u = \sqrt{x}$, then $x = u^2$ and $dx = 2u\, du$
When $x = 0$: $u = 0$; when $x = \pi^2$: $u = \pi$

$$\int_0^{\pi^2} \sin(\sqrt{x})\, dx = \int_0^\pi \sin(u) \cdot 2u\, du = 2\int_0^\pi u \sin(u)\, du$$

Using integration by parts with $v = u$, $dw = \sin(u)du$, so $dv = du$, $w = -\cos(u)$:

$$= 2\left[-u\cos(u)\Big|_0^\pi + \int_0^\pi \cos(u)\, du\right]$$
$$= 2\left[-\pi\cos(\pi) + 0\cos(0) + \sin(u)\Big|_0^\pi\right]$$
$$= 2\left[-\pi \cdot (-1) + 0 + \sin(\pi) - \sin(0)\right]$$
$$= 2\left[\pi + 0 - 0\right] = 2\pi$$

Answer: $\boxed{2\pi}$

4. **(4 points)** Let the function f be given by

$$f(x, y) = \sqrt{x - e^{4y}}.$$

Find an equation of the tangent plane to the graph of f at the point $(5, 0, 2)$.

Solution:

Find partial derivatives:

$$\frac{\partial f}{\partial x} = \frac{1}{2\sqrt{x - e^{4y}}} \cdot 1 = \frac{1}{2\sqrt{x - e^{4y}}}$$
$$\frac{\partial f}{\partial y} = \frac{1}{2\sqrt{x - e^{4y}}} \cdot (-4e^{4y}) = \frac{-4e^{4y}}{2\sqrt{x - e^{4y}}}$$

At $(5, 0)$:

$$f_x(5,0) = \frac{1}{2\sqrt{4}} = \frac{1}{4}$$
$$f_y(5,0) = \frac{-4 \cdot 1}{2\sqrt{4}} = \frac{-4}{4} = -1$$

The tangent plane equation is:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$
$$z - 2 = \frac{1}{4}(x - 5) + (-1)(y - 0)$$
$$z = \frac{1}{4}x - \frac{5}{4} - y + 2$$
$$z = \frac{1}{4}x - y + \frac{3}{4}$$

Answer: $\boxed{z = \frac{1}{4}x - y + \frac{3}{4}}$ or equivalently $\boxed{x - 4y - 4z + 3 = 0}$

5. **(2+2 = 4 points)** Let C be the circle with center $(1, -5)$ that is described by the polar equation

$$r^2 - 2r\cos\theta + ar\sin\theta = -20,$$

where $a \in \mathbb{R}$ is a constant.

(a) Find a.

(b) Find the radius of circle C.

Solution:

(a) Convert the polar equation to Cartesian using $x = r\cos\theta$, $y = r\sin\theta$, $r^2 = x^2 + y^2$:

$$x^2 + y^2 - 2x + ay = -20$$
$$(x^2 - 2x) + (y^2 + ay) = -20$$

Complete the square:

$$\left(x^2 - 2x + 1\right) - 1 + \left(y^2 + ay + \frac{a^2}{4}\right) - \frac{a^2}{4} = -20$$
$$(x-1)^2 + \left(y + \frac{a}{2}\right)^2 = -20 + 1 + \frac{a^2}{4} = -19 + \frac{a^2}{4}$$

The center is at $\left(1, -\frac{a}{2}\right)$. Given that the center is $(1, -5)$:

$$-\frac{a}{2} = -5 \implies a = 10$$

Answer: $\boxed{a = 10}$

(b) With $a = 10$:

$$(x-1)^2 + (y+5)^2 = -19 + \frac{100}{4} = -19 + 25 = 6$$

The radius is $r = \sqrt{6}$.

Answer: $\boxed{\sqrt{6}}$

6. **(4 points)** Show that the limit

$$\lim_{(x,y)\to(0,0)} \frac{2x(x+y)}{x^2+y^2}$$

does **not** exist.

Solution:

We'll approach $(0,0)$ along different paths and show we get different limits.

Path 1: Along $y = x$

$$\lim_{x\to 0} \frac{2x(x+x)}{x^2+x^2} = \lim_{x\to 0} \frac{4x^2}{2x^2} = 2$$

Path 2: Along $y = -x$

$$\lim_{x\to 0} \frac{2x(x-x)}{x^2+x^2} = \lim_{x\to 0} \frac{0}{2x^2} = 0$$

Since we obtain limit 2 along $y = x$ but limit 0 along $y = -x$, the limit does not exist.

Conclusion: The limit does not exist because different paths yield different values (2 versus 0).

7. **(4 points)** Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a differentiable function with

$$\frac{\partial f}{\partial x}(x,y) = 2x \quad \text{and} \quad \frac{\partial f}{\partial y}(x,y) = 3y.$$

If $x(t) = t^2$ and $y(t) = \ln t$, compute $\dfrac{d}{dt} f(x(t), y(t))$ for $t > 0$.

Solution:

Using the multivariable chain rule:

$$\frac{d}{dt} f(x(t), y(t)) = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}$$

Calculate derivatives:

$$\frac{dx}{dt} = 2t$$
$$\frac{dy}{dt} = \frac{1}{t}$$

Substitute:

$$\begin{aligned}\frac{d}{dt} f(x(t), y(t)) &= 2x(t) \cdot 2t + 3y(t) \cdot \frac{1}{t} \\ &= 2(t^2) \cdot 2t + 3(\ln t) \cdot \frac{1}{t} \\ &= 4t^3 + \frac{3\ln t}{t}\end{aligned}$$

Answer: $\boxed{\dfrac{d}{dt} f(x(t), y(t)) = 4t^3 + \dfrac{3\ln t}{t}}$

8. **(2+2 = 4 points)** Consider the function

$$f(x,y) = \frac{1}{3}x^3 + y^2 + 2xy - 6x - 3y + 4.$$

(a) Find all critical points of f.

(b) Determine the absolute minimum of f on the line segment defined by $x = 0$ and $-1 \leq y \leq 1$.

Solution:

(a) Find partial derivatives:

$$f_x = x^2 + 2y - 6$$
$$f_y = 2y + 2x - 3$$

Set both equal to zero:

$$\begin{cases} x^2 + 2y - 6 = 0 \\ 2x + 2y - 3 = 0 \end{cases}$$

From the second equation: $y = \frac{3-2x}{2}$
Substitute into the first:

$$x^2 + 2 \cdot \frac{3-2x}{2} - 6 = 0$$
$$x^2 + 3 - 2x - 6 = 0$$
$$x^2 - 2x - 3 = 0$$
$$(x-3)(x+1) = 0$$

So $x = 3$ or $x = -1$.
For $x = 3$: $y = \frac{3-6}{2} = -\frac{3}{2}$
For $x = -1$: $y = \frac{3+2}{2} = \frac{5}{2}$

Answer: Critical points are $\boxed{(3, -\frac{3}{2})}$ and $\boxed{(-1, \frac{5}{2})}$

(b) On the constraint $x = 0$:

$$f(0,y) = y^2 - 3y + 4$$

This is a function of y alone on $[-1, 1]$.

$$\frac{d}{dy} f(0,y) = 2y - 3 = 0 \implies y = \frac{3}{2}$$

Since $\frac{3}{2} \notin [-1,1]$, the minimum occurs at an endpoint.
Evaluate at endpoints:

$$f(0,-1) = 1 + 3 + 4 = 8$$
$$f(0,1) = 1 - 3 + 4 = 2$$

Answer: The absolute minimum is $\boxed{f(0,1) = 2}$

9. **(4 points)** Find the solution $y = y(x)$ to the following initial value problem:

$$\begin{cases} x^2y' + 2xy = \cos x, \quad x > 0, \\ y(\pi) = 1. \end{cases}$$

Solution:

Divide by x^2:

$$y' + \frac{2}{x}y = \frac{\cos x}{x^2}$$

This is a first-order linear ODE. The integrating factor is:

$$\mu(x) = e^{\int \frac{2}{x}dx} = e^{2\ln|x|} = e^{\ln x^2} = x^2$$

Multiply the equation by x^2:

$$x^2y' + 2xy = \cos x$$

The left side is $\frac{d}{dx}[x^2y]$:

$$\frac{d}{dx}[x^2y] = \cos x$$

Integrate:

$$x^2y = \int \cos x\, dx = \sin x + C$$

Therefore:

$$y = \frac{\sin x + C}{x^2}$$

Apply initial condition $y(\pi) = 1$:

$$1 = \frac{\sin\pi + C}{\pi^2} = \frac{0 + C}{\pi^2}$$

Thus $C = \pi^2$.

Answer: $\boxed{y(x) = \dfrac{\sin x + \pi^2}{x^2}}$

10. **(5 points)** Find the six complex solutions z (in polar form) of the following equation:

$$z^6 - 2z^3 - 3 = 0.$$

Solution:

Let $w = z^3$. Then:

$$w^2 - 2w - 3 = 0$$

Factoring:

$$(w - 3)(w + 1) = 0$$

So $w = 3$ or $w = -1$.

Case 1: $z^3 = 3$

In polar form: $3 = 3e^{i\cdot 0}$

The three cube roots are:

$$z_k = \sqrt[3]{3} \cdot e^{i(0+2\pi k)/3} = 3^{1/3} e^{i\cdot 2\pi k/3}, \quad k = 0, 1, 2$$

- $z_0 = 3^{1/3} e^{i\cdot 0} = 3^{1/3}$
- $z_1 = 3^{1/3} e^{i\cdot 2\pi/3}$
- $z_2 = 3^{1/3} e^{i\cdot 4\pi/3}$

Case 2: $z^3 = -1$

In polar form: $-1 = e^{i\pi}$

The three cube roots are:

$$z_k = e^{i(\pi+2\pi k)/3}, \quad k = 0, 1, 2$$

- $z_3 = e^{i\pi/3}$
- $z_4 = e^{i\pi}$
- $z_5 = e^{i\cdot 5\pi/3}$

Answer: The six solutions in polar form are:

$$\boxed{z_0 = 3^{1/3}, \quad z_1 = 3^{1/3} e^{i\cdot 2\pi/3}, \quad z_2 = 3^{1/3} e^{i\cdot 4\pi/3}}$$

$$\boxed{z_3 = e^{i\pi/3}, \quad z_4 = e^{i\pi}, \quad z_5 = e^{i\cdot 5\pi/3}}$$

11. **(4 points)** Find the solution $y = y(x)$ to the following initial value problem:

$$y'' + 2y' - 3y = 0, \quad y(0) = -1, \quad y'(0) = 0$$

Solution:

The characteristic equation is:

$$r^2 + 2r - 3 = 0$$

Factoring:

$$(r + 3)(r - 1) = 0$$

So $r_1 = -3$ and $r_2 = 1$.

The general solution is:

$$y(x) = c_1 e^{-3x} + c_2 e^x$$

Apply initial condition $y(0) = -1$:

$$y(0) = c_1 + c_2 = -1$$

The derivative is:

$$y'(x) = -3c_1e^{-3x} + c_2e^x$$

Apply initial condition $y'(0) = 0$:

$$y'(0) = -3c_1 + c_2 = 0$$

From the second equation: $c_2 = 3c_1$

Substitute into the first equation:

$$c_1 + 3c_1 = -1 \implies 4c_1 = -1 \implies c_1 = -\frac{1}{4}$$

Then:

$$c_2 = 3 \cdot \left(-\frac{1}{4}\right) = -\frac{3}{4}$$

Answer: $\boxed{y(x) = -\frac{1}{4}e^{-3x} - \frac{3}{4}e^x}$ or equivalently $\boxed{y(x) = -\frac{1}{4}(e^{-3x} + 3e^x)}$